Catalan's Constant

The number that would not be pinned down

A gentle introduction for humans by one, LN McCulloch-James & AN Other · 7 October 2026

Meet the number

Catalan's constant, written G and worth about 0.9159655942, is what you get by adding the reciprocals of the odd squares with alternating signs:

$$\mathrm{G} = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \frac{1}{9^2} - \cdots = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2}.$$

It looks like a harmless cousin of two sums you may already have met, and the comparison is where the mystery begins. Keep the alternating signs but drop the squares and you get Leibniz's series, which equals $\pi/4$; keep the squares but make every sign positive and you get the sum of the reciprocal odd squares, which equals $\pi^2/8$, a close relative of Euler's famous result that the reciprocals of all the squares sum to $\pi^2/6$:

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}, \qquad 1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots = \frac{\pi^2}{8}.$$

Both neighbours collapse into tidy expressions involving π, yet G, sitting exactly between them, has resisted every attempt to write it in closed form since Eugène Catalan studied it in 1865.

That resistance leaves a much more basic question open, one you can state with nothing beyond GCSE vocabulary: is G a fraction? For over a century and a half nobody could say, and a preprint dated 24 September 2026 claims to have finally answered it, proving that G is irrational.

Where it turns up

G would be a curiosity if it lived only in that series, but it keeps surfacing in places that seem to have nothing to do with one another[1].

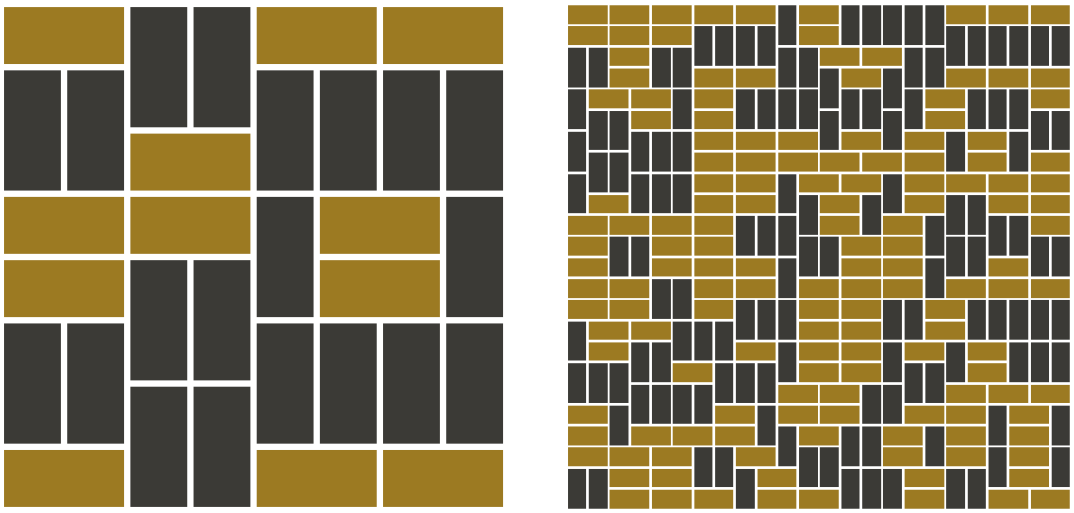
Under a curve. Year 13 students can derive it themselves. Expand $\arctan x$ as its Maclaurin series, divide every term by x, and integrate from 0 to 1; each power x^{2j} integrates to $1/(2j+1)$, which combines with the $1/(2j+1)$ already present to give the odd squares one by one:

$$\frac{\arctan x}{x} = 1 - \frac{x^2}{3} + \frac{x^4}{5} - \cdots \quad \Longrightarrow \quad \int_0^1 \frac{\arctan x}{x}\,dx = 1 - \frac{1}{3^2} + \frac{1}{5^2} - \cdots = \mathrm{G}.$$

[1] For example in the shape of space: The complement of the Whitehead link, a simple two-component knotted loop, carries a hyperbolic geometry whose volume is exactly $4\mathrm{G} \approx 3.664$, and Ian Agol proved in 2010 that this is the smallest volume any orientable hyperbolic three-manifold with two cusps can have. If the new proof stands, that minimum volume is irrational too.

The same number appears as minus the area under $\ln(\tan\theta)$ between 0 and $\pi/4$, which is the form the new paper starts from.

On a chessboard. An ordinary 8×8 board can be covered by 32 dominoes in exactly 12 988 816 different ways, and as the board grows the number of coverings grows like $e^{G/\pi}$ per square. The left-hand picture below shows one of those coverings. Physicists know this problem as molecules settling on a surface, and the section on molecules takes the calculation all the way to Catalan's series.



Left: one of the 12 988 816 ways to cover a chessboard with 32 dominoes. Right: a random covering of a 24×24 patch, a picture of a saturated layer of molecules on a surface. Gold dominoes lie east–west and grey ones north–south; both were drawn by the Monte Carlo routine in the companion script `dimer_catalan.py`, which reaches a random covering by repeatedly rotating pairs of parallel dominoes.

Why can't decimal digits settle irrationality?

A number is *rational* if it can be written as a fraction a/b of whole numbers, and *irrational* otherwise. Every fraction has a decimal expansion that eventually repeats, such as $\frac{3}{7} = 0.428571\,428571\ldots$, and every eventually repeating decimal is a fraction, so a number is irrational exactly when its decimal expansion never settles into a repeating block.

That characterisation cannot be tested by computing digits. Catalan's constant begins $0.9159655942\ldots$, but the ten digits shown are themselves the fraction $\frac{9159655942}{10^{10}}$, and for any number of digits there are fractions agreeing with G that far and differing afterwards. A fraction can also repeat with a period far longer than any computation could reach; $\frac{1}{p}$ for a large prime p can have a repeating block nearly p digits long. Digits can show that a number is close to a fraction, but never that it is not one.

The partial sums of the series show how quickly digits accumulate, and how little they decide. Because the terms alternate in sign and shrink, the error after stopping is always smaller than the first term left out, so stopping after N terms leaves an error below $1/(2N+1)^2$.

```
G = 0.915965594177219015  # reference value, for comparison only
```

```
for N in [1, 10, 100, 1000]:
    s = sum((-1)**j / (2*j + 1)**2 for j in range(N))
    first_omitted = 1 / (2*N + 1)**2
    print(f"N={N:5d} sum={s:.10f} err={abs(G-s):.2e} bound={first_omitted:.2e}")
```

The format `:.2e` prints in scientific notation, so the error and its bound can be compared by eye. A thousand terms give about six correct decimal places, and every one of the partial sums is itself a fraction.

Try it

Q1. Write the partial sum $1 - \frac{1}{9} + \frac{1}{25}$ as a single fraction in lowest terms, and use the alternating-series rule to say how far, at most, it is from G.

Q2. Explain why $0.\overline{142857}$ is rational by multiplying it by 10^6 and subtracting.

The history of irrationality proofs is a history of long waits, because each number defined by an infinite process has needed its own argument.

Number	Irrationality proved	By
$\sqrt{2}$	antiquity	Pythagorean school
e	1737	Euler
π	1761	Lambert
$\zeta(3) = 1 + \frac{1}{2^3} + \frac{1}{3^3} + \cdots$	1978	Apéry
G	claimed, 2026	under examination (see the next two sections)

What "irrational" does and does not say. Irrationality is the first of several increasingly strong statements a number can satisfy. A number is *transcendental* if it is not a root of any polynomial equation with whole-number coefficients; $\sqrt{2}$ is irrational but not transcendental, since it solves $x^2 = 2$, while Lindemann proved in 1882 that π is transcendental. Euler's Basel result $\zeta(2) = 1 + \frac{1}{4} + \frac{1}{9} + \cdots = \frac{\pi^2}{6}$ shows that $\zeta(2)$ is a fraction times π^2, and so it inherits transcendence from π.

Number	Irrational?	Transcendental?	A fraction times π^2?
$3/4$	no	no	no
$\sqrt{2}$	yes	no	no
π	yes (1761)	yes (1882)	no
$\zeta(2) = \pi^2/6$	yes	yes	yes
G	claimed (2026)	unknown	unknown

For G, the new claim reaches only the first column. Whether G might secretly equal a fraction times π^2, as $\zeta(2)$ does, and whether it is transcendental, both remain open.

Extension Linear independence and L-values

Mathematicians phrase these questions in terms of *linear independence over the rationals*. Saying that G is irrational is the same as saying that no equation $A\mathrm{G} + B = 0$ holds with whole numbers A, B not both zero, so that 1 and G are linearly independent. Ruling out $\mathrm{G} = \frac{a}{b}\pi^2$ would need the stronger statement that no equation $A + B\mathrm{G} + C\pi^2 = 0$ holds, so that 1, G and π^2 are linearly independent. The paper's construction deliberately cancels every π^2 term and so says nothing about that stronger question.

Catalan's constant belongs to a family of numbers called Dirichlet L-values: $\mathrm{G} = L(2, \chi_{-4})$, where χ_{-4} is the pattern of signs $+, 0, -, 0, +, 0, -, 0, \ldots$ attached to $1, 2, 3, 4, \ldots$, which repeats every 4 (its *conductor*). Changing the pattern to one that repeats every 3, namely $+, -, 0, +, -, 0, \ldots$, gives the value $L(2, \chi_{-3})$, and in 2024 Calegari, Dimitrov and Tang proved that 1, π^2 and $L(2, \chi_{-3})$ are linearly independent. The corresponding statement for G remains open.

What would a proof need to construct?

Every classical irrationality proof rests on one property of fractions, and a single example shows it. Suppose $x = \frac{3}{7}$. For any whole numbers A and B,

$$A \cdot \frac{3}{7} - B = \frac{3A - 7B}{7},$$

and the numerator $3A - 7B$ is a whole number. Either it is 0, or its size is at least 1, in which case the whole expression has size at least $\frac{1}{7}$.

A	B	$3A - 7B$	$A \cdot \frac{3}{7} - B$
1	0	3	$\frac{3}{7}$
2	1	-1	$-\frac{1}{7}$
5	2	1	$\frac{1}{7}$
7	3	0	0
12	5	1	$\frac{1}{7}$

However cleverly A and B are chosen, the expression $Ax - B$ can never land strictly between $-\frac{1}{7}$ and $\frac{1}{7}$ unless it is exactly zero. The same reasoning applies to any fraction: if $x = a/b$, then $Ax - B = (Aa - Bb)/b$ is either 0 or at least $1/b$ in size.

An irrational number behaves differently. For $\sqrt{2}$, good choices of A and B make $A\sqrt{2} - B$ nonzero and as small as we like:

(A, B)	$(5, 7)$	$(12, 17)$	$(29, 41)$	$(70, 99)$	$(169, 239)$
$A\sqrt{2} - B$	0.0711	-0.0294	0.0122	-0.0051	0.0021

If $\sqrt{2}$ were a/b, these values could never drop below $1/b$ without being zero, and they are never zero, because $A\sqrt{2} = B$ would make $\sqrt{2}$ a fraction with denominator A. Since the values keep shrinking, no denominator b can work.

This turns into a strategy. To prove that a number x is irrational, construct whole numbers A_n and B_n such that the quantities $A_n x - B_n$ are never zero and tend to zero. If x were a/b they would all have size at least $1/b$, so they could not tend to zero, which is a contradiction.

A proof that works: e. The number $e = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots$ shows the strategy succeeding. Take $A_n = n!$ and let B_n be $n!$ times the first $n+1$ terms, $B_n = n!\left(1 + \frac{1}{1!} + \cdots + \frac{1}{n!}\right)$, which is a whole number because $n!/k!$ is a whole number for $k \le n$. What remains is $n!$ times the tail of the series:

$$\begin{aligned} A_n e - B_n &= \frac{n!}{(n+1)!} + \frac{n!}{(n+2)!} + \cdots = \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \cdots \\ &< \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots = \frac{1}{n}, \end{aligned}$$

the last step being the sum of a geometric series with first term and ratio both $\frac{1}{n+1}$. This is positive, so never zero, and it is less than $\frac{1}{n}$, so it tends to zero. If e were a/b, choosing $n \ge b$ would make $A_n e - B_n$ positive and smaller than $\frac{1}{b}$, which the fraction property forbids. So e is irrational.

Three things to establish. Every proof of this kind has to establish three things.

1. **Controlled denominators.** The quantities must be whole-number combinations of x and 1, or at least have denominators that are known and not too large, so that the fraction property gives a fixed minimum size.
2. **Smallness.** The quantities must become smaller than that minimum.
3. **Nonvanishing.** The quantities must never be exactly zero.

For e all three are short. For G the obvious attempt fails on the second, as the partial sums show. Writing the partial sum after N terms as a fraction B/A in lowest terms, the ratio B/A approaches G, but the denominator A grows so quickly that $AG - B$ grows instead of shrinking:

terms N	2	3	4	5	6
partial sum B/A	$\frac{8}{9}$	$\frac{209}{225}$	$\frac{10016}{11025}$	$\frac{91369}{99225}$	$\frac{10956424}{12006225}$
$AG - B$	0.24	−2.9	82.5	−482.3	40865.0

Much cleverer approximations, built by Zudilin and others over the past twenty-five years, have the same defect in a milder form: their ratios close in on G beautifully, but after the denominators are cleared, the whole-number combinations still do not tend to zero.

Try it

Q3. For $x = \frac{2}{5}$, find whole numbers A and B making $Ax - B = \frac{1}{5}$, and explain why no choice makes $Ax - B = \frac{1}{10}$.

Q4. Using the table above, explain in a sentence why the partial sums of Catalan's series cannot by themselves prove that G is irrational.

How would we check that construction?

The new paper meets the three requirements by replacing the single quantity $AG - B$ with a far larger object. It builds square arrays of numbers, each entry an integral chosen so that it equals

a combination of 1, G and π^2 with fractions as coefficients, and then arranges the integrals so that every π^2 cancels. If G were a fraction, each array's determinant would then be a fraction too, and the argument checks the three requirements for those determinants.

1. **Denominators** are controlled by working through the primes one at a time and bounding how many times each prime can divide the denominator. This gives a minimum size the determinants would have to keep if G were a fraction.
2. **Smallness** comes from writing each determinant as an integral over many variables and bounding that integral, which shows that the determinants are in fact smaller than that minimum.
3. **Nonvanishing** is checked by computer, through a finite calculation of Gaussian elimination with whole numbers modulo the prime 101, which shows that the determinants are nonzero for an unending sequence of sizes, enough to keep the argument going.

The two bounds turn out to be incompatible, so G cannot be a fraction.

Checking such an argument is itself a substantial job. Some parts are finite calculations that a computer can repeat exactly, such as the elimination modulo 101 and the rational arithmetic behind the bounds. Others are long chains of estimates over dozens of pages that only expert readers can audit, and a single unjustified step can break the whole argument. Asking which pieces you would trust a computer to verify, and which you would want a person to read, opens a conversation about what a proof is for and whom it has to convince.

The status of the claims. The paper discussed here was written by an unreleased OpenAI model posted on 24 September 2026 as one of 722 manuscripts in a public repository, whose README warns that results without a formal Lean proof could contain errors. As of October 2026 it has not been refereed or formally verified.

A separate proof by Zhi-Wei Sun, posted on 3 September 2026, uses a different construction, and the OpenAI paper states that it uses no result from it. Sun's argument has since received a published technical note, by Dane Wachs, reporting that its derivation does not justify the key estimate it relies on, so that proof is incomplete as written; the note draws no conclusion about whether G is irrational. The two manuscripts stand or fall separately, and a problem in one says nothing directly about the other.

Extension The numbers behind the contradiction

The determinants Δ_N come from matrices of size $n \times n$ with $n = 48N$, and the paper measures their size by

$$L_N = \frac{\ln |\Delta_N|}{n^2} - \tfrac{1}{2} \ln 2.$$

If G were a fraction, the prime-by-prime accounting of the denominators would force $L_N \geq -2.29084$ for large N along the sequence of nonzero determinants, while the integral estimate gives $L_N \leq -2.29094$ for all large N. The two constants agree to four significant figures differing in fifth, so a slightly weaker estimate on either side would have left argument with nothing to say.

Molecules on a surface: where the series comes from

The physical problem. Picture a crystal face whose atoms form a square grid of adsorption sites, and a gas of diatomic molecules, such as O_2 or N_2, each of which sticks lying flat so that it covers two neighbouring sites. When the surface is saturated every site is covered exactly once, and a layer of adsorbed molecules is then precisely a domino tiling of the grid, with each domino standing for one molecule lying either east–west or north–south. Fowler and Rushbrooke posed the counting question in 1937, because the entropy of the layer follows from Boltzmann's formula

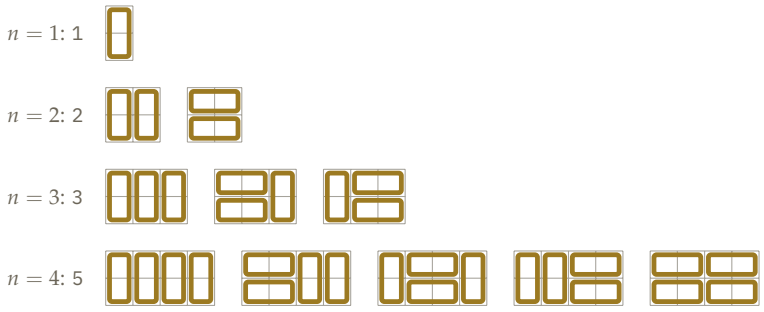
$$S = k_B \ln W,$$

where W is the number of distinct arrangements, and in this idealised model, with no energy preference between orientations, W is simply the number of tilings $T(m,n)$ of an $m \times n$ patch. The quantity of physical interest is the entropy per site in the limit of a large surface,

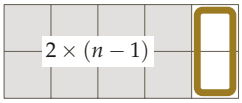
$$s = \lim_{m,n\to\infty} \frac{\ln T(m,n)}{mn}.$$

The calculation has nine steps, and it starts somewhere familiar. Steps 1 to 3 count tilings of a strip two squares high, where the answer is the Fibonacci sequence and the counting matrix is the complex Fibonacci matrix. Step 4 widens the strip to a whole board, and steps 5 and 6 use eigenvectors to turn the count into a product over standing waves on the grid, which can be regrouped as a stack of strips, each growing at its own metallic ratio in the way the Fibonacci strip grows at the golden ratio; step 7 averages over those waves, and steps 8 and 9 are where Catalan's series finally appears. No individual tiling corresponds to an individual term $\pm 1/(2k+1)^2$; the series belongs to the average, and the steps below show how the count reaches it.

Step 1: tiling a strip gives Fibonacci. Take a strip two squares high and n squares long, and write T_n for the number of ways to tile it with dominoes. Drawing every tiling for the first few lengths gives $1, 2, 3, 5$:



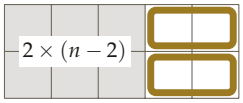
To see why the pattern continues, look at the square in the top right-hand corner of the strip. Some domino must cover it, and there are only two candidates, because that square has only two neighbours, the square below it and the square to its left.



(a) the corner square is covered by a vertical domino



(b) by a horizontal domino: the square below it can now only pair to its left



(c) so the last two columns hold two horizontals

In case (a) a vertical domino fills the whole last column, and what remains is a $2 \times (n-1)$ strip, which can be tiled in T_{n-1} ways. In case (b) a horizontal domino covers the top of the last two columns, and now the bottom right-hand square has lost its upper neighbour, so the only square left for it to pair with is the one on its left; a second horizontal domino is forced underneath the first, and what remains is a $2 \times (n-2)$ strip, which can be tiled in T_{n-2} ways. Every tiling falls into exactly one of these cases, so

$$T_n = T_{n-1} + T_{n-2}, \qquad T_1 = 1,\ T_2 = 2,$$

and the counts run $1, 2, 3, 5, 8, 13, \ldots$, the Fibonacci numbers.

Step 2: a matrix that lists the dominoes. Colour the strip like a chessboard. Each column then contains one white square and one black square, which we label W_c and B_c for column c, and every domino covers one white and one black square. A vertical domino in column c joins W_c to B_c. A horizontal domino joins two squares in neighbouring columns, and because colours alternate along a row it always joins a white square in one column to the black square in the next, so it joins W_c to B_{c+1} or W_{c+1} to B_c. No white square touches anything else.

Now build a matrix K with one row for each white square and one column for each black square, both listed in column order, and let the entry in row W_r and column B_c be the weight of the domino joining them, or 0 if they do not touch. Giving vertical dominoes the weight v and horizontal dominoes the weight h, the description above says that the diagonal entries are v, the entries immediately beside the diagonal are h, and everything else is 0. Such a matrix is called *tridiagonal*:



the 2×4 strip

$$K_4 = \begin{array}{c} \\ W_1 \\ W_2 \\ W_3 \\ W_4 \end{array} \begin{array}{c} \begin{array}{cccc} B_1 & B_2 & B_3 & B_4 \end{array} \\ \begin{pmatrix} v & h & 0 & 0 \\ h & v & h & 0 \\ 0 & h & v & h \\ 0 & 0 & h & v \end{pmatrix} \end{array}$$

The determinant of a square matrix is a sum of products, one product for every way of choosing exactly one entry from each row with no column used twice, each product carrying a sign $+$ or $-$. For K, choosing one entry from each row means giving every white square a black partner, and using no column twice means no black square is used twice, so the nonzero products are exactly the tilings, each recorded as the product of its dominoes' weights.

The smallest case shows both the idea and the difficulty. For the 2×2 board the matrix is

$$K_2 = \begin{pmatrix} v & h \\ h & v \end{pmatrix}, \qquad \det K_2 = v \cdot v - h \cdot h,$$

where $v \cdot v$ picks W_1B_1 and W_2B_2, the tiling by two vertical dominoes, and $h \cdot h$ picks W_1B_2 and W_2B_1, the tiling by two horizontal ones. The matrix records each domino's orientation by the position of its entry, on the diagonal or beside it, but the determinant subtracts the second tiling from the first, so with $v = h = 1$ it gives $1 - 1 = 0$ although the board has two tilings.

A sum with every sign set to $+$ would count correctly. That sum has a name, the *permanent*, and for a 2×2 matrix it is $ad + bc$ where the determinant is $ad - bc$; it was introduced by Cauchy

in 1812, and in 1979 Leslie Valiant proved that computing permanents is as hard as a whole class of counting problems for which no fast method is believed to exist. A determinant, by contrast, can be found in roughly N^3 steps by row reduction, so the aim is to choose v and h so that the cheap determinant does the permanent's work.

Step 3: the complex Fibonacci matrix. Write $D_n = \det K_n$. Expanding a 3×3 case along its last column shows the pattern:

$$D_3 = \det \begin{pmatrix} v & h & 0 \\ h & v & h \\ 0 & h & v \end{pmatrix} = v \det \begin{pmatrix} v & h \\ h & v \end{pmatrix} - h \det \begin{pmatrix} v & h \\ 0 & h \end{pmatrix} = v\,D_2 - h \cdot h\,D_1.$$

The bottom entry v multiplies the determinant of the top-left block, which is D_2. The entry h above it multiplies the determinant left after deleting its row and column, and that smaller matrix has only h in its last row, so it equals $h\,D_1$. The same two moves work for every n, because the last column of K_n always holds exactly two nonzero entries, v at the bottom and h just above it. The bottom entry multiplies the top-left block, which is K_{n-1}; the entry h carries a minus sign, since its row and column numbers, $n-1$ and n, add to an odd number, and deleting its row and column leaves a matrix whose last row contains a single h in the corner, so that expanding along that row leaves h times the top-left block K_{n-2}. This gives the recurrence

$$D_n = v\,D_{n-1} - h^2\,D_{n-2}.$$

Read as tilings this is Step 1 again. The term $v\,D_{n-1}$ pairs W_n with B_n, the vertical domino closing the strip in case (a); the term $h^2 D_{n-2}$ pairs W_n with B_{n-1} and W_{n-1} with B_n, the two horizontal dominoes forced together in cases (b) and (c). The only difference from $T_n = T_{n-1} + T_{n-2}$ is the minus sign, and the weights decide what that sign does:

	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8
$v = 1$, $h = 1$, so $h^2 = 1$	1	0	−1	−1	0	1	1	0
$v = 1$, $h = i$, so $h^2 = -1$	1	2	3	5	8	13	21	34

With every weight equal to 1 the recurrence $D_n = D_{n-1} - D_{n-2}$ runs around a cycle of period six and counts nothing. With $h = i$ the horizontal pair contributes $i \cdot i = -1$, the minus sign turns that into $+1$, and the recurrence becomes $D_n = D_{n-1} + D_{n-2}$, which is Fibonacci exactly. The matrix is then the complex Fibonacci matrix,

$$K_n = \begin{pmatrix} 1 & i & 0 & \cdots & 0 \\ i & 1 & i & \ddots & \vdots \\ 0 & i & 1 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & i \\ 0 & \cdots & 0 & i & 1 \end{pmatrix}, \qquad K_2 = \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}, \qquad \det K_2 = 1 - i^2 = 2,$$

and $\det K_n = T_n$ for every n.

Only the product matters. The recurrence never sees h on its own. The term $-h^2 D_{n-2}$ comes from one entry just above the diagonal multiplied by one just below it, so the determinant

records only the product of those two entries, and any other pair with the same product gives the same sequence. Multiplying the rows of K_n by $1, d, d^2, \ldots, d^{n-1}$ and dividing its columns by the same numbers leaves the determinant alone, since every product in the determinant uses each row and each column exactly once and the factors cancel; it multiplies each entry below the diagonal by d and divides each entry above it by d, so every above-and-below product is unchanged. Taking $d = -i$ turns the complex Fibonacci matrix into a real one, with -1 above the diagonal and $+1$ below:

$$\begin{pmatrix} 1 & i & 0 \\ i & 1 & i \\ 0 & i & 1 \end{pmatrix} \quad \longrightarrow \quad \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad \text{both with determinant } 3.$$

Read as dominoes, the real version gives a horizontal domino the weight $+1$ when its white square lies to the right of its black square and -1 when it lies to the left, so the sign depends on which way the domino points along the row from black to white. The two matrices count equally well, and the passage from one to the other is a change of bookkeeping that the determinant cannot detect.

This freedom is developed in general in *The Characteristic Equation*. For a tridiagonal matrix with 1 on the diagonal, b above it and c below, the determinants satisfy $D_n = D_{n-1} - bc\, D_{n-2}$, the characteristic equation is $x^2 = x - bc$, and the product bc is the one quantity that the rescalings of rows and columns cannot change. With $bc = -1$ the equation is $x^2 = x + 1$, the golden ratio's, whichever of the pairs (i, i), $(-1, 1)$ or $(1, -1)$ is used; with $bc = +1$ it is $x^2 = x - 1$, whose roots lie on the unit circle, and that is the period-six row of the table above. The complex weight i is kept for the full board because it treats left and right alike, so that a single rule serves every row without deciding which way each domino points.

Step 4: from a strip to a board. On a full $m \times n$ board, give every horizontal neighbour the weight i and every vertical neighbour the weight 1, exactly as on the strip; this is Kasteleyn's choice of 1961, and the strip shows why it is the natural one. To see that every tiling still contributes equally, use the move from the Monte Carlo picture below: any tiling of a rectangle can be turned into any other by repeatedly rotating a 2×2 block holding two parallel dominoes, a fact proved by William Thurston in 1990. Such a flip changes the product of weights in that block from $i \times i = -1$ to $1 \times 1 = 1$, and it also swaps the black partners of the block's two white squares, a single swap that reverses the sign the determinant attaches to the term. The two sign changes cancel, so the flip leaves the term unchanged, which is the same cancellation that turned $D_{n-1} - h^2 D_{n-2}$ into a sum on the strip. Since flips connect every tiling to every other, all the terms are equal, each has modulus 1, and

$$|\det K| = T(m, n).$$

Step 5: eigenvectors, eigenvalues and determinants. Step 4 gives the count as a determinant, but the matrix for a large board has thousands of rows, and expanding it term by term is hopeless. The way through is to find the directions in which the matrix acts most simply. A matrix M turns a vector into another vector, usually pointing in a new direction; an *eigenvector* is a nonzero vector $\mathbf{v}$ whose direction survives, so that the matrix only rescales it,

$$M\mathbf{v} = \lambda \mathbf{v},$$

and the scale factor λ is the corresponding *eigenvalue*. The complex Fibonacci matrix K_2 makes a good first example, because its eigenvectors can be checked by hand:

$$\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+i \\ 1+i \end{pmatrix} = (1+i) \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

$$\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1-i \\ -(1-i) \end{pmatrix} = (1-i) \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

So $(1,1)$ has eigenvalue $1+i$ and $(1,-1)$ has eigenvalue $1-i$. Physicists will recognise these two vectors as the in-phase and anti-phase motions of a pair of coupled pendulums, the two *normal modes* in which the whole system moves together at a single frequency.

Two vectors are *linearly independent* when neither is a multiple of the other, and more generally a set of vectors is linearly independent when none of them can be built from the others. In two dimensions any two independent vectors form a *basis*, meaning that every vector can be written as a combination of them in exactly one way; for these two,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{x+y}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{x-y}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

and in N dimensions any N independent vectors do the same job. When a matrix has a basis made of eigenvectors, its whole action is simple to describe: split any vector into its eigenvector parts, rescale each part by its own eigenvalue, and add the parts back together.

This explains why the determinant is the product of the eigenvalues. A 2×2 determinant measures the factor by which the matrix scales areas, and a 3×3 determinant the factor by which it scales volumes. Measured in the eigenvector basis, the matrix stretches the first direction by λ_1, the second by λ_2, and so on, so areas and volumes are scaled by $\lambda_1 \lambda_2 \cdots \lambda_N$. For K_2,

$$\lambda_1 \lambda_2 = (1+i)(1-i) = 1 - i^2 = 2 = \det K_2, \qquad \lambda_1 + \lambda_2 = (1+i) + (1-i) = 2 = 1 + 1,$$

and the second equation illustrates a companion fact: the *trace* of a matrix, the sum of its diagonal entries, equals the sum of its eigenvalues. With complex eigenvalues the stretching picture also involves rotation and is harder to draw, but the algebra is the same, and both facts hold for every square matrix.

Step 6: the board's eigenvectors are standing waves. To use Step 5 we need a square matrix whose determinant is tied to $\det K$. Build the matrix A with one row and one column for every square of the board, entry i between horizontal neighbours, 1 between vertical neighbours, and 0 elsewhere. For the 2×2 board, listing the squares in the order W_1, W_2, B_1, B_2 gives

$$A = \left(\begin{array}{cc|cc} 0 & 0 & 1 & i \\ 0 & 0 & i & 1 \\ \hline 1 & i & 0 & 0 \\ i & 1 & 0 & 0 \end{array}\right) = \begin{pmatrix} 0 & K \\ K^{\mathsf{T}} & 0 \end{pmatrix}.$$

The top-left block is zero because white squares never touch each other on a chessboard, the bottom-right block is zero for the same reason, the top-right block is K, and the bottom-left block is K^{T}, the transpose of K with rows and columns exchanged, because touching is symmetric.

Swapping the top half of the rows with the bottom half turns A into $\left(\begin{smallmatrix} K^{\mathsf{T}} & 0 \\ 0 & K \end{smallmatrix}\right)$; each row swap only changes the sign of a determinant, the determinant of a block-diagonal matrix is the product of the blocks' determinants, and transposing does not change a determinant, so

$$\det A = \pm \det K^{\mathsf{T}} \det K = \pm(\det K)^2, \qquad \text{so} \qquad |\det A| = T(m,n)^2.$$

The same argument works on every board, so it remains to find the eigenvectors of A.

Start with a single row of n squares, where the neighbour matrix P sends a vector $\mathbf{v} = (v_1, \ldots, v_n)$ to the vector whose x-th entry is $v_{x-1} + v_{x+1}$, with $v_0 = v_{n+1} = 0$ because there are no squares beyond the ends. For three squares,

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$P \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} = \sqrt{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}, \qquad P \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad P \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix} = -\sqrt{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}.$$

Plotted against position, these three eigenvectors are the first three standing waves on a string fixed at both ends: a single hump, then two humps of opposite sign with a node at the centre, then three humps alternating in sign. Their eigenvalues are $\sqrt{2}$, 0 and $-\sqrt{2}$; the trace of P is 0, which matches their sum, and $\det P = 0$, which matches their product. A row of three squares cannot be tiled by dominoes, and the zero eigenvalue is how the determinant records it.

The eigenvalues for a row of any length can be found before the eigenvectors, by the method of Step 3. A number λ is an eigenvalue of P exactly when $P - \lambda I$ sends some nonzero vector to zero, which happens exactly when $\det(\lambda I - P) = 0$. The matrix $\lambda I - P$ is tridiagonal, with λ on the diagonal and -1 on either side, so the expansion of Step 3, with $v = \lambda$ and $h = -1$, shows that its determinant E_n satisfies

$$E_n = \lambda E_{n-1} - E_{n-2}, \qquad E_0 = 1,\ E_1 = \lambda,$$

whose characteristic equation $x^2 - \lambda x + 1 = 0$ has two roots multiplying to 1. Writing $\lambda = 2\cos a$ with $0 < a < \pi$, the roots are $\cos a \pm i \sin a = e^{\pm ia}$, and the same steps that give Binet's formula for the Fibonacci numbers give

$$E_n = \frac{e^{i(n+1)a} - e^{-i(n+1)a}}{e^{ia} - e^{-ia}} = \frac{\sin\big((n+1)a\big)}{\sin a},$$

which vanishes exactly when $(n+1)a$ is a multiple of π. That gives n different eigenvalues, $2\cos\big(\pi k/(n+1)\big)$ for $k = 1, \ldots, n$, and since E_n is a polynomial of degree n in λ there can be no others. The period-six row of the table in Step 3 is this same sequence at $\lambda = 1$, where $a = \pi/3$, and its zeros at $n = 2, 5, 8, \ldots$ say that 1 is an eigenvalue of P exactly when $n+1$ is a multiple of 3.

The eigenvectors follow. Try $v_x = \sin(xa)$ with $a = \pi k/(n+1)$ for $k = 1, \ldots, n$. The identity $\sin(x-1)a + \sin(x+1)a = 2\cos a \, \sin xa$ gives

$$v_{x-1} + v_{x+1} = 2\cos\left(\frac{\pi k}{n+1}\right) v_x,$$

and the end conditions hold because $\sin 0 = 0$ and $\sin(\pi k) = 0$. These are the standing waves on the row, with k counting the humps; for $n = 3$ they are exactly the vectors above, since $\sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$ and $\sin\frac{\pi}{2} = 1$. The n waves are linearly independent because they belong to different eigenvalues: if some combination $c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ were zero, applying $(P - \lambda_2 I)(P - \lambda_3 I)\cdots(P - \lambda_n I)$ would remove every term but the first and leave $c_1(\lambda_1 - \lambda_2)\cdots(\lambda_1 - \lambda_n)\,\mathbf{v}_1 = \mathbf{0}$, so that $c_1 = 0$, and the same argument with the roles exchanged removes every other coefficient. So the waves form a basis, and splitting a vector into this basis is the discrete version of a Fourier series, which writes any shape of a string as a sum of its standing waves. This is why the eigenvectors are often called *modes*, and the eigenvalues, $2\cos(\pi k/(n+1))$, play the part of the modes' frequencies.

On the full board, the eigenvectors are products of a wave along the rows and a wave up the columns, like the vibration patterns of a rectangular drum skin:

$$v_{x,y} = \sin\left(\frac{\pi k x}{n+1}\right)\sin\left(\frac{\pi j y}{m+1}\right).$$

At each square, A adds i times the values at the left and right neighbours and 1 times the values above and below. The horizontal sum changes only the first factor, multiplying by $2\cos(\pi k/(n+1))$, and the vertical sum changes only the second, multiplying by $2\cos(\pi j/(m+1))$, so

$$A\mathbf{v} = \lambda_{jk}\mathbf{v}, \qquad \lambda_{jk} = 2\cos\frac{\pi j}{m+1} + 2i\cos\frac{\pi k}{n+1}, \qquad |\lambda_{jk}|^2 = 4\cos^2\frac{\pi j}{m+1} + 4\cos^2\frac{\pi k}{n+1}.$$

There are mn of these drum modes, one for each pair (j,k), and their eigenvalues are all different, since the real part of λ_{jk} fixes j and the imaginary part fixes k; the argument used for the row therefore shows that they are linearly independent, and so they form a basis for the board. By Step 5, $|\det A|$ is the product of the moduli $|\lambda_{jk}|$, so $T^2 = \prod|\lambda_{jk}|$, and taking one more square root gives Kasteleyn's formula,

$$T(m,n) = \prod_{j=1}^{m}\prod_{k=1}^{n}\left(4\cos^2\frac{\pi j}{m+1} + 4\cos^2\frac{\pi k}{n+1}\right)^{1/4}.$$

The strip closes the circle. With $m = 2$ both values of $4\cos^2(\pi j/3)$ equal 1, so $T(2,n) = \prod_k\left(1 + 4\cos^2\frac{\pi k}{n+1}\right)^{1/2}$; for $n = 3$ the factors are 3, 1 and 3, giving $\sqrt{9} = 3$, and the next paragraph shows why the cosine product reproduces the Fibonacci numbers of Step 1 for every n.

The board as a stack of metallic strips. Kasteleyn's formula treats rows and columns alike, but the count can also be organised one direction at a time, and doing so brings the complex Fibonacci matrix back. Fix a column wave and leave the shape along the row free, taking

$$v_{x,y} = u_x \sin\frac{\pi j y}{m+1}.$$

The vertical neighbours multiply the second factor by $\mu_j = 2\cos\left(\pi j/(m+1)\right)$, and the horizontal neighbours add $i(u_{x-1} + u_{x+1})$ to the first, so on vectors of this shape A acts on the row $(u_1,\dots,u_n)$ as the $n \times n$ matrix

$$M_n(\mu_j) = \begin{pmatrix} \mu_j & i & & \\ i & \mu_j & \ddots & \\ & \ddots & \ddots & i \\ & & i & \mu_j \end{pmatrix},$$

which is the complex Fibonacci matrix of Step 3 with its diagonal 1 replaced by μ_j. In *The Characteristic Equation* these are the *metallic* matrices: by the expansion of Step 3 their determinants satisfy

$$D_n(\mu) = \mu\, D_{n-1}(\mu) + D_{n-2}(\mu), \qquad D_0 = 1,\ D_1 = \mu,$$

with characteristic equation $x^2 = \mu x + 1$, so that $\mu = 1$ gives the Fibonacci numbers and the golden ratio, $\mu = 2$ gives the Pell numbers and the silver ratio $1 + \sqrt{2}$, and so on through the metals. Each of the m column waves picks out a strip of this kind, the strips do not interact, and in the basis of column waves A falls apart into m separate blocks, so that its determinant is the product of theirs and

$$T(m,n)^2 = \Big| \prod_{j=1}^{m} D_n(\mu_j) \Big|.$$

The board is therefore a stack of m metallic strips whose metals μ_j all lie strictly between -2 and 2, so that every strip sits somewhere between the trivial case $\mu = 0$ and the silver ratio, with the golden strip $\mu = 1$ appearing whenever $m + 1$ is a multiple of 3.

Three checks show the formula at work. With $m = 2$ the metals are $\mu = \pm 1$; the strip with $\mu = 1$ is the complex Fibonacci matrix itself, so $D_n(1) = T_n$, and the sequence $(-1)^n T_n$ satisfies the recurrence with $\mu = -1$, so the product is T_n^2 and $T(2,n) = T_n$ for every n, which is Step 1 recovered from the eigenvalues. With $m = 3$ and $n = 2$ the metals are $\sqrt{2}$, 0 and $-\sqrt{2}$, the determinants $D_2(\mu) = \mu^2 + 1$ are 3, 1 and 3, and the product 9 gives $T(3,2) = 3$, the three tilings of a 2×3 rectangle met in Step 1. When m is odd the middle wave has $\mu = 0$, where the recurrence collapses to $D_n(0) = D_{n-2}(0)$ with values $1, 0, 1, 0, \ldots$, and so $T(m,n) = 0$ whenever m and n are both odd, as it must be, since dominoes always cover an even number of squares.

The complex Fibonacci matrix's own eigenvalues. Since $K_n = I + iP$, every row wave is also an eigenvector of K_n, with eigenvalue $1 + 2i\cos\left(\pi k/(n+1)\right)$, so the eigenvalues of the complex Fibonacci matrix all lie on the vertical line $\operatorname{Re}\lambda = 1$, between $1 - 2i$ and $1 + 2i$, as the eigenvalue plotter in the incidence-matrix app shows. Pairing k with $n + 1 - k$ pairs each eigenvalue with its complex conjugate, and since $(1 + iy)(1 - iy) = 1 + y^2$, the count of Step 1 becomes a product of real factors,

$$T_n = \det K_n = \prod_{k=1}^{n} \Big(1 + 2i\cos\frac{\pi k}{n+1}\Big) = \prod_{k<(n+1)/2} \Big(1 + 4\cos^2\frac{\pi k}{n+1}\Big),$$

where for odd n the unpaired middle eigenvalue is 1 and changes nothing. The first few cases need surprisingly different numbers:

n	eigenvalues of K_n	numbers needed	T_n
2	$1 \pm i$	i	2
3	$1,\ 1 \pm \sqrt{2}\,i$	$i,\ \sqrt{2}$	3
4	$1 \pm \varphi i,\ 1 \pm \varphi^{-1} i$	$i,\ \sqrt{5}$	$(1+\varphi^2)(1+\varphi^{-2}) = 5$
5	$1,\ 1 \pm i,\ 1 \pm \sqrt{3}\,i$	$i,\ \sqrt{3}$	$1 \cdot 2 \cdot 4 = 8$

The golden ratio turns up for $n = 4$ because $2\cos(\pi/5) = \varphi$, and the eigenvalues $1 \pm i$ of K_2 turn up again inside K_5. The second observation is general. If $d + 1$ divides $n + 1$, every angle $\pi k/(d+1)$ is also one of the angles $\pi k'/(n+1)$, so the eigenvalues of K_d all reappear among those of K_n, and T_d divides T_n, which is the eigenvalue form of the classical fact that

F_{d+1} divides F_{n+1} whenever $d+1$ divides $n+1$. When $n+1$ is prime nothing smaller reappears, and the cosines become harder to write down: K_6 already needs $\cos(\pi/7)$, a root of the cubic $8x^3 - 4x^2 - 4x + 1 = 0$.

Try it

Q5. A row of two squares has neighbour matrix $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Show that $(1,1)$ and $(1,-1)$ are eigenvectors, find their eigenvalues, and check that the product of the eigenvalues equals $\det P$. How many ways are there to tile a row of two squares, and how does $|\det P|$ record it?

Q6. The $4 \times n$ board has metals $\mu_j = 2\cos(\pi j/5)$, which are $\pm\frac{1+\sqrt{5}}{2}$ and $\pm\frac{\sqrt{5}-1}{2}$. Use $D_2(\mu) = \mu^2 + 1$ to show that $T(4,2) = 5$, and check the answer against Step 1.

Q7. Which of the earlier counts $T_1, T_2, \ldots, T_{10}$ must divide $T_{11} = 144$, according to the eigenvalue argument? Check each one.

Step 7: from a product to an integral. Kasteleyn's formula is exact, but it is a product of mn factors, and the entropy needs its behaviour as the board grows without limit. Logarithms turn the product into a sum, and for a large board that sum becomes an integral, in the same way that a sum of thin rectangles becomes the area under a curve.

A one-variable warm-up shows the mechanism. Take the n angles $\theta_k = \pi k/(n+1)$ for $k = 1, \ldots, n$, which are spread evenly across $(0,\pi)$ with spacing $\pi/(n+1)$, and average a function over them. Multiplying each value by the spacing gives a sum of rectangle areas, which approaches $\int_0^\pi f(\theta)\,d\theta$, so the plain average approaches the integral divided by the length of the interval:

$$\frac{1}{n}\sum_{k=1}^{n} f\left(\frac{\pi k}{n+1}\right) \longrightarrow \frac{1}{\pi}\int_0^\pi f(\theta)\,d\theta \qquad \text{as } n \to \infty.$$

For $f(\theta) = \cos^2\theta$ both sides can be found exactly. The integral gives $\frac{1}{\pi}\int_0^\pi \cos^2\theta\,d\theta = \frac{1}{2}$, while the sum turns out to equal $\frac{n-1}{2n}$; for three angles, $\pi/4$, $\pi/2$ and $3\pi/4$, the values of $\cos^2$ are $\frac{1}{2}$, 0 and $\frac{1}{2}$, whose average is $\frac{1}{3} = \frac{3-1}{2\cdot 3}$. As n grows, $\frac{n-1}{2n} = \frac{1}{2} - \frac{1}{2n}$ approaches $\frac{1}{2}$, the gap shrinking like $1/n$.

The tiling count works the same way in two directions at once. Taking logarithms of Kasteleyn's formula and dividing by the number of squares gives

$$\frac{\ln T(m,n)}{mn} = \frac{1}{4}\cdot\frac{1}{m}\sum_{j=1}^{m}\frac{1}{n}\sum_{k=1}^{n}\ln\left(4\cos^2\frac{\pi j}{m+1} + 4\cos^2\frac{\pi k}{n+1}\right),$$

which is a quarter of the average of $\ln|\lambda_{jk}|^2$ over all the standing waves. Each average becomes an integral, $\frac{1}{m}\sum_j \to \frac{1}{\pi}\int_0^\pi d\theta$ and $\frac{1}{n}\sum_k \to \frac{1}{\pi}\int_0^\pi d\phi$, so

$$s = \frac{1}{4\pi^2}\int_0^\pi\int_0^\pi \ln\left(4\cos^2\theta + 4\cos^2\phi\right)\,d\phi\,d\theta.$$

Since $\cos^2(\pi-\theta) = \cos^2\theta$, the graph of $\cos^2$ is symmetric about $\theta = \pi/2$, so each integral over $(0,\pi)$ is twice the integral over $(0,\pi/2)$. Folding both directions multiplies by 4, which cancels the $\frac{1}{4}$:

$$s = \frac{1}{\pi^2}\int_0^{\pi/2}\int_0^{\pi/2} \ln\left(4\cos^2\theta + 4\cos^2\phi\right)\,d\phi\,d\theta.$$

The convergence can be watched directly, using the code at the end of this section:

board	8×8	40×40	200×200	1000×1000	limit
$\ln T/(\text{squares})$	0.2559	0.2842	0.2901	0.2913	0.2916

The approach is slow because squares along the edge of a board have fewer neighbours, and on an $n\times n$ board the edge squares make up a fraction of roughly $4/n$ of the whole.

Try it

Q8. Check the formula $\frac{n-1}{2n}$ for the average of $\cos^2\left(\frac{\pi k}{n+1}\right)$ when $n=1$ and when $n=2$.

Q9. Use Kasteleyn's formula to count the tilings of the 2×2 board. (Each of the four factors is the same.)

Step 8: the inner integral. The double integral is done one variable at a time, ϕ first with θ held fixed. The result is

$$I(\theta)=\int_0^{\pi/2}\ln\left(4\cos^2\theta+4\cos^2\phi\right)d\phi=\pi\,\operatorname{arsinh}(\cos\theta),$$

where arsinh is the inverse of the hyperbolic sine $\sinh x=\frac{1}{2}(e^x-e^{-x})$, so that $\operatorname{arsinh} y=\ln\left(y+\sqrt{y^2+1}\right)$.

The metallic strips give a derivation that stays within A-level. Like the Fibonacci numbers, each $D_n(\mu)$ is a combination of the nth powers of the two roots of its characteristic equation $x^2=\mu x+1$,

$$r=\tfrac{1}{2}\left(\mu+\sqrt{\mu^2+4}\right),\qquad r'=\tfrac{1}{2}\left(\mu-\sqrt{\mu^2+4}\right),\qquad D_n(\mu)=\frac{r^{n+1}-r'^{\,n+1}}{r-r'},$$

and since $rr'=-1$, for $\mu>0$ the root r exceeds 1 while r' lies between -1 and 0, so the powers of r' fade and $D_n(\mu)$ grows like r^n, in the way that F_{n+1}/F_n settles on the golden ratio. Putting $\mu=2\cos\theta$ gives $r=\cos\theta+\sqrt{\cos^2\theta+1}$, and the logarithmic form of arsinh shows at once that

$$\ln r=\operatorname{arsinh}(\cos\theta),$$

so the right-hand side of the formula is π times the logarithm of a metallic ratio; at $\theta=\pi/3$, where $\mu=1$, it is $\ln\varphi$ for the golden ratio φ. The left-hand side comes from computing the same determinant the other way. The metallic matrix is $M_n(\mu)=\mu I+iP$, so by Step 6 its eigenvalues are $\mu+2i\cos\left(\pi k/(n+1)\right)$, and the determinant, the product of the eigenvalues, gives

$$\frac{1}{n}\ln|D_n(\mu)|^2=\frac{1}{n}\sum_{k=1}^{n}\ln\left(4\cos^2\theta+4\cos^2\frac{\pi k}{n+1}\right).$$

As n grows the left-hand side approaches $2\ln r=2\operatorname{arsinh}(\cos\theta)$, by the growth just described, while by the warm-up of Step 7 the right-hand side approaches $\frac{1}{\pi}\int_0^{\pi}\ln(4\cos^2\theta+4\cos^2\phi)\,d\phi$, which is $\frac{2}{\pi}I(\theta)$ after folding. Equating the two limits gives $I(\theta)=\pi\operatorname{arsinh}(\cos\theta)$.

Read through the strips, the whole calculation says that the entropy per site is half the average, over all the column waves, of the logarithm of each strip's metallic ratio. A board made only

of golden strips would have entropy $\frac{1}{2}\ln\varphi \approx 0.2406$ per site, which is exactly the value for the $2 \times n$ strip of Step 1; the full board averages over strips of every metal between 0 and 2, and the average comes out at 0.2916.

The extension box below gives the classical route to the same formula, differentiating under the integral sign, and the formula can also be tested directly. At $\theta = \pi/2$, where $\cos\theta = 0$, the right-hand side is $\pi\,\operatorname{arsinh}(0) = 0$, and the left-hand side is

$$I(\tfrac{\pi}{2}) = \int_0^{\pi/2} \ln(4\cos^2\phi)\,d\phi = 2\int_0^{\pi/2} \ln(2\cos\phi)\,d\phi = \pi\ln 2 + 2\int_0^{\pi/2} \ln(\cos\phi)\,d\phi.$$

Euler found the last integral with a neat trick. Call it L. Replacing ϕ by $\frac{\pi}{2} - \phi$ shows that $\int_0^{\pi/2}\ln(\sin\phi)\,d\phi$ is also L, so

$$2L = \int_0^{\pi/2} \ln(\sin\phi\cos\phi)\,d\phi = \int_0^{\pi/2} \ln(\tfrac{1}{2}\sin 2\phi)\,d\phi = -\frac{\pi}{2}\ln 2 + \int_0^{\pi/2} \ln(\sin 2\phi)\,d\phi,$$

and the substitution $u = 2\phi$, together with the symmetry of $\sin u$ about $u = \pi/2$, shows that the final integral is L again. Hence $2L = -\frac{\pi}{2}\ln 2 + L$, so $L = -\frac{\pi}{2}\ln 2$, and $I(\frac{\pi}{2}) = \pi\ln 2 - \pi\ln 2 = 0$, as the formula predicts. At other angles a numerical integration does the checking:

θ	0	0.5	1.0	$\pi/2$
$I(\theta)$ by numerical integration	2.768917	2.488490	1.624096	0.000000
$\pi\,\operatorname{arsinh}(\cos\theta)$	2.768917	2.488490	1.624096	0

Substituting $I(\theta)$ into the double integral leaves a single integral,

$$s = \frac{1}{\pi^2}\int_0^{\pi/2} \pi\,\operatorname{arsinh}(\cos\theta)\,d\theta = \frac{1}{\pi}\int_0^{\pi/2} \operatorname{arsinh}(\cos\theta)\,d\theta.$$

Extension Where the inner integral comes from

Write $4\cos^2\phi = 2 + 2\cos 2\phi$ and substitute $\psi = 2\phi$, so that $I(\theta) = \frac{1}{2}J(a)$ with

$$J(a) = \int_0^{\pi} \ln(a + 2\cos\psi)\,d\psi, \qquad a = 2 + 4\cos^2\theta \geq 2.$$

Treat a as a variable and differentiate under the integral sign, which is allowed here because the integrand is smooth in a:

$$J'(a) = \int_0^{\pi} \frac{d\psi}{a + 2\cos\psi}.$$

The substitution $t = \tan(\psi/2)$, with $\cos\psi = \frac{1-t^2}{1+t^2}$ and $d\psi = \frac{2\,dt}{1+t^2}$, turns this into a standard arctangent integral,

$$J'(a) = \int_0^{\infty} \frac{2\,dt}{(a+2) + (a-2)t^2} = \frac{2}{\sqrt{(a+2)(a-2)}}\cdot\frac{\pi}{2} = \frac{\pi}{\sqrt{a^2-4}}.$$

Integrating back with respect to a gives $J(a) = \pi\ln\left(\frac{a+\sqrt{a^2-4}}{2}\right) + C$. For very large a the integrand is close to $\ln a$, so $J(a) \approx \pi\ln a$, and the formula also gives about $\pi\ln a + C$; hence $C = 0$. Finally, define u by $\cosh 2u = a/2 = 1 + 2\cos^2\theta$. The identity $\cosh 2u = 1 + 2\sinh^2 u$ gives $\sinh u = \cos\theta$, and $\sqrt{a^2-4} = 2\sinh 2u$, so

$$\frac{a+\sqrt{a^2-4}}{2} = \cosh 2u + \sinh 2u = e^{2u}, \qquad J(a) = 2\pi u, \qquad I(\theta) = \pi u = \pi\,\operatorname{arsinh}(\cos\theta).$$

Try it

Q10. Show that $\int_0^{\pi/2} \ln(\sin\phi)\,d\phi = \int_0^{\pi/2} \ln(\cos\phi)\,d\phi$ by substituting $\phi = \frac{\pi}{2} - \psi$.

Q11. Show that $\operatorname{arsinh}\frac{1}{2} = \ln\frac{1+\sqrt{5}}{2}$, so that the golden strip, $\mu = 1$, has metallic ratio $e^{\operatorname{arsinh}(1/2)} = \varphi$.

Step 9: the series appears. Two ingredients are needed: a series for $\operatorname{arsinh} x$ and the integrals of the powers of $\cos\theta$. Both come from A-level techniques.

The series. The derivative of $\operatorname{arsinh} x$ is $(1+x^2)^{-1/2}$, whose binomial expansion, valid for $|x| \le 1$, is

$$(1+x^2)^{-1/2} = 1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 - \frac{5}{16}x^6 + \cdots.$$

Integrating term by term, and using $\operatorname{arsinh} 0 = 0$, gives

$$\operatorname{arsinh} x = x - \frac{1}{6}x^3 + \frac{3}{40}x^5 - \frac{5}{112}x^7 + \cdots.$$

Only odd powers appear because the derivative contains only even powers, and the signs alternate because the binomial expansion of a negative power alternates.

The integrals. Substituting $x = \cos\theta$ means we need $\int_0^{\pi/2} \cos^{2k+1}\theta\,d\theta$ for $k = 0, 1, 2, \ldots$. Writing $\cos^{2k+1}\theta = (1-\sin^2\theta)^k \cos\theta$ and substituting $u = \sin\theta$ makes each one a polynomial integral:

$$\begin{aligned}
\int_0^{\pi/2} \cos\theta\,d\theta &= 1,\\
\int_0^{\pi/2} \cos^3\theta\,d\theta &= \int_0^1 (1-u^2)\,du = 1 - \frac{1}{3} = \frac{2}{3},\\
\int_0^{\pi/2} \cos^5\theta\,d\theta &= \int_0^1 (1-u^2)^2\,du = 1 - \frac{2}{3} + \frac{1}{5} = \frac{8}{15}.
\end{aligned}$$

Multiplying. Integrating the series term by term, each coefficient multiplies the matching integral:

power	coefficient in $\operatorname{arsinh} x$	$\int_0^{\pi/2} \cos^{\text{power}}\theta\,d\theta$	product
x^1	1	1	1
x^3	$-\frac{1}{6}$	$\frac{2}{3}$	$-\frac{2}{18} = -\frac{1}{9}$
x^5	$\frac{3}{40}$	$\frac{8}{15}$	$\frac{24}{600} = \frac{1}{25}$

The odd squares and alternating signs of Catalan's series emerge one term at a time, $1 - \frac{1}{9} + \frac{1}{25} - \cdots$, and Question 12 asks you to confirm the next term, $-\frac{1}{49}$. The pattern continues for ever. In general the coefficient of x^{2k+1} is $(-1)^k \frac{(2k)!}{4^k(k!)^2(2k+1)}$ and the integral of $\cos^{2k+1}\theta$ is $\frac{4^k(k!)^2}{(2k+1)!}$; multiplying them, the factors $4^k(k!)^2$ cancel, and $\frac{(2k)!}{(2k+1)!} = \frac{1}{2k+1}$ supplies a second copy of the odd number already present, so each product is $\frac{(-1)^k}{(2k+1)^2}$. One odd number comes from the arsinh series and the other from the cosine integral. Therefore

$$\int_0^{\pi/2} \operatorname{arsinh}(\cos\theta)\,d\theta = 1 - \frac{1}{9} + \frac{1}{25} - \frac{1}{49} + \cdots = \mathrm{G}, \qquad s = \frac{\mathrm{G}}{\pi} \approx 0.2916.$$

(Integrating an infinite series term by term needs justification in general, and here it comes from the size of the coefficients. The coefficient of x^{2k+1} has size $\frac{1}{2k+1}\cdot\frac{(2k)!}{4^k(k!)^2}$, and the second factor, which equals $\frac{1}{2}\cdot\frac{3}{4}\cdots\frac{2k-1}{2k}$, is at most $1/\sqrt{2k+1}$, as squaring it and comparing each factor $\frac{2j-1}{2j}$ with $\frac{2j}{2j+1}$ makes the product telescope. So for every x between 0 and 1 the kth term is at most $(2k+1)^{-3/2}$, these bounds have a finite sum, and the error from stopping the series after K terms is below the same tail $\sum_{k>K}(2k+1)^{-3/2}$ for every θ at once; after integrating over an interval of length $\frac{\pi}{2}$ the error is below $\frac{\pi}{2}$ times that tail, which tends to zero.)

Try it

Q12. Show that $\int_0^{\pi/2}\cos^7\theta\,d\theta=\frac{16}{35}$, and multiply by the coefficient $-\frac{5}{112}$ to obtain the fourth term of Catalan's series.

What the number means. The entropy per site is $0.2916\,k_B$, and since each molecule covers two sites the entropy per adsorbed molecule is $2\mathrm{G}/\pi\approx 0.583\,k_B$. A tempting first guess treats each molecule as free to choose between two orientations, which would give $\ln 2\approx 0.693\,k_B$ per molecule, and the shortfall between 0.693 and 0.583 measures how much the requirement to pack the surface completely, with no overlaps and no gaps, restricts the molecules' freedom; the right-hand picture on page 2 shows what a typical saturated layer looks like. The model is an idealisation, since real layers are rarely saturated and orientations rarely cost exactly the same energy, and the partially filled version, in which single atoms share the surface with the molecules, has never been solved exactly.

Extension Dominoes and trees

The incidence-matrix app builds, for any network, the Laplacian $L=D-A$, where D lists how many neighbours each point has and A is the adjacency matrix, and it demonstrates Kirchhoff's Matrix-Tree theorem: deleting one row and the matching column of L leaves a matrix whose determinant counts the *spanning trees*, the ways of choosing edges that join every point without closing a loop. Temperley noticed in 1974 that these trees are domino tilings in disguise. Draw an $a\times b$ grid of points and enlarge it to a board of $(2a-1)\times(2b-1)$ squares, so that the points become the squares in odd rows and odd columns, the edges become the squares between two points, and the faces become the remaining squares; then remove the corner square that stands for one of the points. In any tiling, the domino covering each remaining point square also covers one neighbouring edge square, which picks out one edge leaving each point, and those edges always form a spanning tree leading towards the missing corner. The counts agree exactly:

grid of points	2×2	2×3	3×3	3×4
spanning trees, by the Matrix-Tree theorem	4	15	192	2415
board, with one corner removed	3×3	3×5	5×5	5×7
domino tilings	4	15	192	2415

The Laplacian of the grid has the same drum modes as Step 6, with eigenvalues $4\sin^2\frac{\pi j}{2a}+4\sin^2\frac{\pi k}{2b}$, sines appearing where the dimer problem had cosines, and the Matrix-Tree theorem turns the tree count into a product of them. Averaging their logarithms as in Step 7 gives the same double integral, now taken over a range four times as large, so the number of spanning trees grows like $e^{4\mathrm{G}/\pi}\approx 3.21$ per point. The factor of four is the bijection again, because the board has about four squares for every point of the grid, and Catalan's constant measures how many ways there are to

connect a large grid without loops just as it measures how many ways there are to cover a surface with molecules.

Extension Two lattices, two L-functions

Kasteleyn's eigenvalues can be read as a polynomial evaluated on the unit circle. Writing $x = e^{i\theta}$ and $y = e^{i\phi}$, the eigenvalue $2\cos\theta + 2i\cos\phi$ of Step 6 is $x + \frac{1}{x} + i\left(y + \frac{1}{y}\right)$, and the entropy per molecule, $2\mathrm{G}/\pi$, is the average of the logarithm of its size over all angles, an average that mathematicians call the *Mahler measure* of the polynomial. The same recipe works on a honeycomb, the grid of hexagons whose dimer coverings are the lozenge tilings that look like stacks of cubes, where the polynomial is $1 + x + y$. Its Mahler measure was found by Chris Smyth in 1981, and it brings in the second constant of the extension on L-values, $L(2, \chi_{-3}) = 1 - \frac{1}{2^2} + \frac{1}{4^2} - \frac{1}{5^2} + \frac{1}{7^2} - \cdots$:

lattice	turns that fix it	number system	polynomial	entropy per molecule
square	quarter turn	Gaussian, $a + bi$	$x + \frac{1}{x} + i\left(y + \frac{1}{y}\right)$	$\frac{2}{\pi}L(2, \chi_{-4}) = \frac{2\mathrm{G}}{\pi} \approx 0.5831$
honeycomb	sixth turn	Eisenstein, $a + b\omega$	$1 + x + y$	$\frac{3\sqrt{3}}{4\pi}L(2, \chi_{-3}) \approx 0.3231$

The pattern of signs in each L-function follows the symmetry of its lattice: the square grid is carried to itself by multiplying by i, a quarter turn, and the triangular grid of hexagon centres by multiplying by the Eisenstein unit $-\omega^2 = e^{i\pi/3}$, a sixth turn.

The Eisenstein half of the table met the news in October 2026. The same public repository that holds the Catalan preprint also holds a claimed proof of the *quasi-Riemann hypothesis*, the statement that the zeros of $\zeta(s)$ stay a fixed distance away from the line $\mathrm{Re}\, s = 1$, and its method works with a whole family of L-functions attached to the Eisenstein integers. Experts reading the shorter of its two versions, which shows that no zeros lie beyond $\mathrm{Re}\, s = 11/12$, broadly expect it to stand, although it has not been refereed. The constants in the table are values at $s = 2$, where every series converges comfortably and the zeros play no part, so the tilings and the new proof share their L-functions while asking different questions of them: how molecules pack says nothing about where the zeros lie, and where the zeros lie says nothing about whether G is a fraction.

Extension When the surface is not full

Let some sites stay empty, and give every empty site a weight z that measures how readily the gas leaves sites bare. If m_k is the number of ways to place k molecules on a board of N sites, the partition function is

$$Z(z) = \sum_k m_k\, z^{N-2k}, \qquad \text{for the } 2 \times 2 \text{ board} \qquad Z(z) = z^4 + 4z^2 + 2,$$

since the 2×2 board holds no molecules in 1 way, one molecule in 4 ways and two in 2 ways. Nobody has a formula for Z on a large board, which is why the partially filled layer has never been solved exactly, but in 1972 Ole Heilmann and Elliott Lieb proved that on every board, and indeed on every network, all the zeros of Z lie on the imaginary axis. For the 2×2 board, $z^2 = -2 \pm \sqrt{2}$, so $z = \pm i\sqrt{2 \pm \sqrt{2}}$. For a single row of n squares the theorem is a fact already met in Step 6: comparing the two sums shows that $E_n(iz) = i^n Z(z)$, so the zeros of Z are $-i$ times the eigenvalues $2\cos\left(\pi k/(n+1)\right)$ of the neighbour matrix, and they are imaginary because a symmetric matrix has real eigenvalues.

The physical consequence is that the layer has no phase transition. A sudden change in a material's behaviour, such as a melting, shows up mathematically as zeros of Z closing in on the positive

values of z that a real gas can have, and zeros confined to the imaginary axis can never do that, so the partially filled layer changes smoothly with z even though its entropy is unknown. Lee and Yang proved a theorem of the same kind for magnets in 1952, with the zeros on a circle. Results like these are why the Riemann hypothesis, which places the zeros of ζ on the line $\operatorname{Re} s = \frac{1}{2}$, has long been suspected of describing some hidden physical system, with the zeros at $\frac{1}{2} + i$ times the eigenvalues of a symmetric operator, an idea associated with Hilbert and Pólya. The complex Fibonacci matrix is a small model of that shape: $K_n = I + iP$ with P symmetric, so every eigenvalue lies on the line $\operatorname{Re} \lambda = 1$. Nobody has found the operator for ζ, and in that picture the quasi-Riemann hypothesis would be the weaker statement that the zeros lie in a strip around a line instead of on it.

Extension An analogy with Euler's constant γ

The harmonic numbers $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n}$ grow like $\ln n$, the area under $1/x$, and what is left over when the continuous approximation is removed settles down to a constant: $H_n - \ln n \to \gamma \approx 0.5772$, the Euler–Mascheroni constant. The tiling count behaves the same way one level up. Evaluating Kasteleyn's formula for even n and fitting the result gives

$$\ln T(n,n) \approx \frac{\mathrm{G}}{\pi}\, n^2 + \left(\frac{2\mathrm{G}}{\pi} - \ln\left(1+\sqrt{2}\right)\right) n + c.$$

The n^2 term is the bulk entropy, playing the part of $\ln n$. The n term, about $-0.298\, n$, is the entropy lost along the edges, where squares have fewer neighbours, and its constant $\ln(1+\sqrt{2}) = \operatorname{arsinh} 1$ is the inner integral of Step 8 evaluated at $\theta = 0$, the edge of the range of modes, which in the language of the metallic strips is the logarithm of the silver ratio $1+\sqrt{2}$, the limiting metal $\mu = 2$ that the column waves approach but never reach. The constant c is what remains once area and edge are removed, the counterpart of γ:

n	20	40	80	160	320	640
remainder	0.103763	0.103446	0.103363	0.103342	0.103336	0.103335

The remainder settles near 0.10333, with each doubling of n cutting the gap by about a factor of four. No closed form for it is offered here.

The analogy carries a warning as well as a pattern. Whether γ is irrational is still unknown, and so is the status of the packing penalty $\ln 2 - 2\mathrm{G}/\pi$, and the tiling limit cannot settle the question for G either. The whole numbers in this problem are the counts $T(n,n)$, and G enters only through their logarithms, so it sits in an exponent; the limit is G/π, which brings in π; and the approach is slow, with an error of about $0.3/n$. None of this produces whole-number combinations $A\mathrm{G} - B$ that shrink to zero, which is what the fraction property needs. Constants that measure the gap between a discrete count and its continuous shadow tend to resist proof, and the successful arguments, Apéry's for $\zeta(3)$ and the claimed one for G, work by building special combinations of that linear kind.

Counting tilings in Python. The function below evaluates Kasteleyn's formula through logarithms, because the product itself overflows a floating-point number by about a 40×40 board, a small lesson in why physicists work with entropy, the logarithm of a count. The second function counts the same tilings exactly, in whole numbers, by running the metallic recurrence for each column wave and multiplying the results. The companion script `dimer_catalan.py` goes further, counting tilings by brute force, building the complex Fibonacci matrix and the full signed matrix, checking the standing-wave eigenvalues, verifying each integral above

numerically, and drawing a random saturated layer.

```
from math import cos, pi, log, exp, sqrt

G = 0.915965594177219015  # reference value, for comparison only

def log_tilings(m, n):
    """Natural log of the number of domino tilings of an m x n board."""
    return 0.25 * sum(log(4*cos(pi*j/(m+1))**2 + 4*cos(pi*k/(n+1))**2)
                      for j in range(1, m+1) for k in range(1, n+1))

def metallic(n, mu):
    """D_n(mu) from D_n = mu*D_(n-1) + D_(n-2), with D_0 = 1, D_1 = mu."""
    a, b = 1, mu
    for _ in range(n - 1):
        a, b = b, mu*b + a
    return b if n > 0 else a

def tilings_by_strips(m, n):
    """T(m, n) as the square root of the product of m metallic strips."""
    p = 1.0
    for j in range(1, m+1):
        p *= metallic(n, 2*cos(pi*j/(m+1)))
    return round(sqrt(abs(p)))

print(round(exp(log_tilings(8, 8))))   # 12988816
print(tilings_by_strips(8, 8))         # 12988816
for n in [8, 40, 200]:
    print(n, log_tilings(n, n) / n**2, G / pi)
```

Answers to the questions

Q1. $1 - \frac{1}{9} + \frac{1}{25} = \frac{225-25+9}{225} = \frac{209}{225}$. The first omitted term is $\frac{1}{49}$, so the partial sum is within $\frac{1}{49} \approx 0.020$ of G; in fact $\frac{209}{225} \approx 0.9289$ and $G \approx 0.9160$.

Q2. If $x = 0.\overline{142857}$ then $10^6 x = 142857.\overline{142857}$, so $999999x = 142857$ and $x = \frac{142857}{999999} = \frac{1}{7}$.

Q3. $A = 3, B = 1$ gives $\frac{6}{5} - 1 = \frac{1}{5}$. In general $Ax - B = \frac{2A-5B}{5}$, a whole number divided by 5, which can be 0 or at least $\frac{1}{5}$ in size but never $\frac{1}{10}$.

Q4. The quantities $AG - B$ built from the partial sums grow in size instead of tending to zero, so the smallness requirement fails and no contradiction with the fraction property can arise.

Q5. $P(1,1) = (1,1)$, so $(1,1)$ has eigenvalue 1; $P(1,-1) = (-1,1) = -1 \cdot (1,-1)$, so $(1,-1)$ has eigenvalue -1. The product is -1, and $\det P = 0 \cdot 0 - 1 \cdot 1 = -1$. A row of two squares has exactly one tiling, a single horizontal domino, and $|\det P| = 1$.

Q6. The metals come in pairs $\pm\mu$ with the same μ^2. For $\mu = \frac{1+\sqrt{5}}{2}$, $\mu^2 = \frac{3+\sqrt{5}}{2}$ and $D_2 = \frac{5+\sqrt{5}}{2}$; for $\mu = \frac{\sqrt{5}-1}{2}$, $\mu^2 = \frac{3-\sqrt{5}}{2}$ and $D_2 = \frac{5-\sqrt{5}}{2}$. The product of the four is $\left(\frac{5+\sqrt{5}}{2} \cdot \frac{5-\sqrt{5}}{2}\right)^2 = 5^2$,

so $T(4,2) = 5$, which is T_4 from Step 1, since a 4×2 board is a 2×4 strip turned on its side.

Q7. The divisors of 12 are $1, 2, 3, 4, 6, 12$, so $d + 1 \in \{2, 3, 4, 6\}$ gives $d = 1, 2, 3, 5$, and $T_1 = 1$, $T_2 = 2$, $T_3 = 3$ and $T_5 = 8$ must divide 144; indeed $144 = 72 \cdot 2 = 48 \cdot 3 = 18 \cdot 8$. None of the others need divide it, and $T_4 = 5$, $T_6 = 13$ and $T_7 = 21$ do not.

Q8. For $n = 1$ the only angle is $\pi/2$, where $\cos^2 = 0$, and $\frac{1-1}{2} = 0$. For $n = 2$ the angles are $\pi/3$ and $2\pi/3$, where $\cos^2 = \frac{1}{4}$, so the average is $\frac{1}{4} = \frac{2-1}{2 \cdot 2}$.

Q9. With $m = n = 2$ every factor is $4 \cdot \frac{1}{4} + 4 \cdot \frac{1}{4} = 2$, so $T = (2^4)^{1/4} = 2$.

Q10. With $\phi = \frac{\pi}{2} - \psi$ we have $\cos\phi = \sin\psi$, $d\phi = -d\psi$, and the limits $0, \frac{\pi}{2}$ become $\frac{\pi}{2}, 0$; reversing the limits removes the minus sign.

Q11. $\operatorname{arsinh} \frac{1}{2} = \ln\left(\frac{1}{2} + \sqrt{\frac{1}{4} + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right) = \ln \frac{1+\sqrt{5}}{2}$, and exponentiating gives φ.

Q12. $\int_0^1 (1 - u^2)^3 \, du = 1 - 1 + \frac{3}{5} - \frac{1}{7} = \frac{16}{35}$, and $-\frac{5}{112} \cdot \frac{16}{35} = -\frac{80}{3920} = -\frac{1}{49}$.

Sources. OpenAI, *Catalan's constant is irrational*, preprint, 24 September 2026, in the repository `https://github.com/openai/math/tree/main/preprints`. Zhi-Wei Sun, *Catalan's constant is irrational*, arXiv:2609.04176, 3 September 2026. Dane Wachs, *A note on a recent claimed proof of the irrationality of Catalan's constant*, arXiv:2609.22339, 2026. L. N. McCulloch-James, *The Characteristic Equation: an invariant beneath the representations of a linear recurrence*, `https://leelemacjames.github.io/Characteristic_Equation.pdf`. L. N. McCulloch-James, *Complex eigenvalue field extensions of Fibonacci matrices*, Dwight School London. L. N. McCulloch-James, *Incidence matrices: from Fibonacci to graph topology*, interactive, `https://leelemacjames.github.io/adjacencyMatrix/`. H. N. V. Temperley, *Enumeration of graphs on a large periodic lattice*, in *Combinatorics*, London Mathematical Society Lecture Note Series 13, 1974. C. J. Smyth, *On measures of polynomials in several variables*, Bulletin of the Australian Mathematical Society 23, 1981. O. J. Heilmann and E. H. Lieb, *Theory of monomer-dimer systems*, Communications in Mathematical Physics 25, 1972. OpenAI, *The quasi-Riemann hypothesis*, preprints of 30 September and 5 October 2026, in the repository above; reported in *Quanta Magazine*, 19 October 2026, `https://www.quantamagazine.org/ai-proved-the-quasi-riemann-hypothesis-what-happens-now-20261019/`.